
Toward collective intelligence: evolutionary pressures for cooperative language models

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1 Abstract

WIP not ready for review Training language models to cooperate with humans and with each other remains an open problem: current alignment methods optimize individual model behavior without accounting for the multi-agent dynamics that govern real-world deployment. We introduce a reinforcement learning framework that translates Nowak’s evolutionary rules of cooperation — direct reciprocity, indirect reciprocity, and group selection — into training pressures for large language models. Our reward function operates at three scales: individual payoff via a general-sum-game, dyadic coordination via an HKB phase-coupling term whose phase transition recovers Nowak’s analytical thresholds, and collective coherence via a prediction-error penalty that makes cheap talk costly. We evaluate on benchmarks that peruses for ability to communicate, coordinate [etc]. We find that [X, Y, Z]. [Ideally: one sentence on the broader implication.]

2 Introduction

Systems that achieve high levels of collective coordination gain substantial advantages over those that cannot: they surpass their individual capabilities and often give rise to entirely new behaviours [Lévy, 1997]. Nature has solved the coordination problem through collective intelligence at every major transition of biological complexity [Prokopenko et al., 2024]: from prokaryotes that through mutualistic interactions became mitochondria, to cleaner fish that cooperate through social memory [Bshary and Grutter, 2006], to human economies that coordinate through price signals [Lyons and Levin, 2024]. Building aligned AI systems with humans requires solving a version of the same problem. In this work we ask whether it is possible to draw inspiration from nature’s evolutionary processes in order to develop a new post-training regime for language models that elicits cooperative capabilities at scale.

To capture these conditions systematically, we ground our work in Nowak’s rules of cooperation [Nowak, 2006], which formalise when cooperation arises in evolutionary biology. The rules are: (*kin selection*), which leverages genetic proximity to promote cooperation; (*direct reciprocity*) through repeated encounters; (*indirect reciprocity*) through reputation; (*network reciprocity*) through interaction topology; and (*group selection*) leveraging inter-group competition. The translation from the biological framework to the context of language models is non-trivial — in the latter, cooperation must be sustained through communication and shared beliefs rather than genetic relatedness. This mapping constitutes part of our contribution, echoing the broader tradition in which each level of organisation captures the essence of the details below it, abstracting them into effective theories that serve cooperation [Anderson, 1972].

We translate these pressures into a reinforcement learning framework with three reward terms: an individual game payoff, a dyadic coordination pressure grounded in the HKB model [Kelso, 2012] whose phase transition recovers Nowak’s analytical threshold for indirect reciprocity, and a collective coherence penalty that makes cheap talk costly. We evaluate on a benchmark sweeping

each mechanism’s key parameter and on the MACHIAVELLI suite [Pan et al., 2023], on Sotopia [], MoToMQA [?], ExploreToM, and Concordia []. We find [results summary].

Contributions. (1) A new mapping from Nowak’s evolutionary rules to training mechanisms for LLMs, grounding each biological rule in a social or cognitive analogue. (2) A multi-scale reward function whose analytical thresholds, inspired by coordination dynamics and the free energy principle, recover Nowak’s predictions. (3) A cooperation benchmark that measures whether a model’s behaviour tracks each mechanism’s key parameter. (4) Empirical evidence that RL fine-tuning with these pressures enables small models to acquire conditional cooperation [TODO: specify results once available].

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3 Related Literature

Direct reciprocity. Several recent studies test LLMs in iterated social dilemmas. [Akata et al., 2025] find that GPT-4 handles self-interested games but struggles with coordination in iterated Prisoner’s Dilemma. [Fontana et al., 2024] report that Llama2 displays stronger cooperation propensity than humans, rarely defecting first. [Tennant et al., 2025] show that standard RL produces defection in iterated PD and that moral reward functions are required to shape cooperative behaviour, explicitly citing Nowak. [Duque et al., 2025] formalise opponent shaping and prove that agents learn tit-for-tat cooperation through advantage alignment.

Indirect reciprocity. [Vallinder and Hughes, 2024] study indirect reciprocity across generations of LLM agents in the Donor Game, finding that cooperation depends heavily on the base model: Claude 3.5 Sonnet sustains it while GPT-4o and Gemini 1.5 Flash largely fail. [Zhu et al., 2026] introduce a gossip-driven framework where open-ended language sustains indirect reciprocity with theoretical guarantees. [Ren et al., 2025] propose a dual-level reputation framework showing emergent cooperative clusters and social isolation of defectors. [Santos et al., 2025] extract social norms from 21 LLMs and evaluate them through evolutionary game theory. More broadly, reputation-based cooperation depends on the honesty of the signals that carry reputational information; theoretical and empirical work shows that widespread dishonest gossip can collapse reputation systems entirely [Számádó et al., 2021].

Network reciprocity. A critical negative finding: [Pereda et al., 2025] directly test Nowak’s condition $b/c > k$ with LLMs and find that static network structure fails to stabilise cooperation — humans cooperate more on structured networks, but GPT-4 does the opposite. [NetworkGames authors, 2025] identify a “Paradox of Connectivity” in which small-world networks allow defection to spread, with the personality of network hubs as the key determinant.

Group selection. [Tonini and Galke, 2025] confirm super-additive cooperation effects when combining intra-group dynamics with inter-group competition. [Piatti et al., 2024] introduce GovSim for common-pool resource management, finding that all but the most powerful LLMs fail at sustainability. [Willis et al., 2025] simulate evolutionary dynamics of LLM populations via Moran processes.

Cooperation as collective intelligence. A parallel literature from developmental biology reframes alignment not as value matching but as the scaling of cognitive light cones across interacting agents [Lyons et al., 2026, Levin, 2019]. In this view, the mechanisms that coordinate cells into organisms — stress sharing, memory anonymisation, distributed credit assignment — are structurally analogous to the mechanisms that coordinate people into economies [Lyons and Levin, 2024]. The price system, bioelectric networks, and reputation systems all function as cognitive glues that expand what agents effectively regulate. Defection, under this framing, is not a strategic pathology but the default mode of an agent whose connection to the collective signalling architecture has weakened: the self shrinks, and events that were once within the agent’s regulatory scope become externalities [Lyons et al., 2026]. This perspective reframes each of Nowak’s rules as a different technology for building cognitive glue, and cooperation failure as a failure to scale the self.

Positioning. No prior work implements all four of Nowak’s mechanisms within a unified training framework for LLMs. Existing studies test individual rules in isolation, typically under explicit

game-theoretic framing that models can exploit directly. We depart from this pattern in three ways: we activate multiple rules simultaneously as training pressures rather than evaluation benchmarks; we ground theory of mind in coordination dynamics rather than explicit prediction; and we connect the resulting framework to the broader programme of understanding cooperation as a problem of scaling collective intelligence across substrates [Lyons et al., 2026, Levin, 2019].

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4 Methods

The framing developed above — cooperation as the scaling of collective intelligence across substrates — suggests that translating Nowak’s biological rules into training pressures for language models requires grounding each rule in a mechanism that expands what an agent effectively regulates. Bioelectric networks, reputation systems, and price signals are all examples of what function as cognitive glue that expands what agents effectively regulate [Lyons and Levin, 2024, Levin, 2019]. Defection, under this framing, is the default mode of an agent whose connection to the collective is weak; in other words, the cognitive boundary of the self does not encompass those the agent is interacting with, and the agent sees events as externalities happening to the other rather than as something with relevant impact to itself [Lyons et al., 2026]. Each of Nowak’s rules can then be read as a different technology for expanding that boundary of a self. This perspective directly motivates our reward design: its three terms operate at the individual, the dyad, and the group level, progressively widening the scale at which the agent’s behaviour is answerable to outcomes.

4.1 Reward design

The reward function combines three of Nowak’s terms: direct reciprocity, indirect reciprocity and group-level selection. As well as capturing the dynamics of our general sum game, the Donor’s Game.

Term 1 provides the immediate strategic incentive, which represents the payoff matrix of the donor’s game. Term 2 provides a metacognitive and theory-of-mind pressure that rewards the agent for coupling to its partner’s behaviour, expanding the understanding of self to the dyad. Finally, Term 3 provides a collective pressure that penalises the group when its members’ stated beliefs diverge from reality, penalising cheap-talk [Crawford and Sobel, 1982] and expanding the dyad to the selfness of the group.

Term 1: Donor’s game environment All experiments use the donor’s game as the base interaction. In each round, a designated donor decides whether to pay a cost $c > 0$ to confer a benefit $b > c$ on a partner. The environment varies the interaction structure: who plays whom, whether reputations are visible, and whether agents form persistent groups, all in order to instantiate each of Nowak’s mechanisms. The raw per-round payoff for agent k at round t is

$$\pi_t^k = b \mathbf{1}\{a_t^{-k} = C\} - c \mathbf{1}\{a_t^k = C\}, \quad (1)$$

where $a_t^k \in \{C, D\}$ is agent k ’s action and a_t^{-k} is the partner’s. Normalised to $[0, 1]$:

$$r_{1,t}^k = \frac{\pi_t^k + c}{b + c}. \quad (2)$$

Term 2: coordination dynamics via the HKB potential. In the introduction we argued that cooperation emerges when autonomous agents expand the boundary of what they can effectively coordinate over. Term 2 operationalises this at the dyadic scale: it rewards the agent for incorporating the partner into its own regulatory dynamics. We do not reward cooperation directly, because doing so would overfit the model to a single strategy. Instead, we reward *coordination*: the capacity to sustain coherent interaction patterns and transition between them as conditions change. The pressure toward cooperation itself comes from Term 1, the game payoff. Term 2 shapes *how* the agent interacts and models its opponent.

Drawing on Kelso’s coordination dynamics [Kelso, 2012, Tognoli et al., 2020] and the metastability framework of Hancock et al. [2024], we represent the agent’s coordination state as a relative phase variable φ evolving in a double-well potential landscape. In the metastable regime, the agent dwells

near contextually appropriate coordination patterns while retaining the capacity to transition when conditions change — without needing to cross energy barriers between strategies [Kelso, 2012].

The coordination variable φ measures the degree of behavioural synchrony between the agent and its current partner. The HKB coupling and bistability constants are tied directly to Nowak’s mechanisms: the coupling strength is set to $4q$, where q is reputation visibility, and the bistability strength is set to c/b , the donor’s-game cost–benefit ratio (both derivations given in *Parameter mappings* below). The extended HKB model then gives the potential:

$$V(\varphi) = -\delta\omega \varphi - 4q \cos(\varphi) - \frac{c}{b} \cos(2\varphi), \quad (3)$$

In this potential, stable coordination states sit at the minima. Since we want the reward to be highest when coordination is strongest, we negate the potential. In the self-play setting considered here, $\delta\omega = 0$ (see *Parameter mappings* below), so the detuning term vanishes and the reward reduces to:

$$r_{2,t}^k = \lambda_{\text{tom}} \left[4q \cos(\varphi_t^k) + \frac{c}{b} \cos(2\varphi_t^k) \right]. \quad (4)$$

Computing φ . Each round, we encode actions as $+1$ (cooperate) or -1 (defect) and compute per-round coordination $s_t = a_t^k \times a_t^{-k}$, so mutual cooperation or mutual defection yields $+1$ (in-phase) while mismatched actions yield -1 (anti-phase). Over a window of W recent rounds with the same partner:

$$\varphi = \frac{\pi(1 - \bar{s})}{2}, \quad \bar{s} = \frac{1}{W} \sum_t s_t. \quad (5)$$

Sustained mutual coordination maps to $\varphi \approx 0$ (in-phase well), stable alternation to $\varphi \approx \pi$ (anti-phase well), and uncoordinated play to $\varphi \approx \pi/2$ (the ridge between wells, which is penalised). The $\cos(2\varphi)$ term creates wells at both 0 and π , rewarding *any* stable coordination pattern, since the goal is to train an agent that can lock into whatever pattern is appropriate given the game conditions. φ is computed per partner: the agent maintains a separate interaction window for each partner it encounters, so coordination history with one partner does not bleed into the reward when playing another.

Memory is organised hierarchically. Within a single game (same partner, N rounds), the coordination window of length W accumulates round-by-round, so φ at round k reflects the W most recent interactions with that partner. At the start of each new game — a new partner, a new scenario — the window is emptied and φ is reinitialised at $\pi/2$ (the metastable regime, corresponding to no prior coordination). No coordination statistics leak across games, consistent with the requirement that theory-of-mind inference operate freshly on each new partner.

Parameter mappings and Nowak alignment. The HKB parameters are chosen so that the potential’s dynamics mirror Nowak’s mechanisms. $\delta\omega$ (detuning) captures the intrinsic frequency difference between the two agents; because we train in self-play, where both sides of the dyad are sampled from the same weights, there is no intrinsic asymmetry and we set $\delta\omega = 0$. A non-zero $\delta\omega$ would only be appropriate when training two distinct policies against each other, where a linear drift term $\delta\omega \cdot \varphi$ in the potential could legitimately reflect a genuine asymmetry between the two sides of the dyad. The coupling strength is set to $4q$, where q is the reputation visibility from Nowak’s indirect reciprocity rule: higher q increases how tightly the agent couples to the coordination dynamics — reputation, in this framing, is a form of cognitive glue that strengthens the connection between agents who have never interacted directly. The factor of 4 is the specific constant that aligns the HKB phase transition with Nowak’s threshold, as derived below. The bistability strength is set to c/b , the donor’s-game cost–benefit ratio, which encodes the tension that pulls agents toward defection.

With these substitutions, the HKB phase transition (which occurs when the bistability-to-coupling ratio equals $1/4$) becomes $(c/b)/(4q) = 1/4$, which simplifies to $q = c/b$ — exactly Nowak’s threshold for indirect reciprocity. The three dynamical regimes (monostable when $q > c/b$, metastable at $q = c/b$, bistable when $q < c/b$) map directly onto Nowak’s prediction for when cooperation, flexibility, or defection should dominate.

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Pressure 3: collective coherence and the cost of cheap talk. Group selection requires individual behaviour to be legible at the group level — agents must be able to form and communicate accurate beliefs about what the group will do. In biological collectives, this legibility is achieved through

interceptive signals: shared physiological states that allow the collective to monitor itself [Lyons et al., 2026]. In purely linguistic systems, however, language is cheap. An agent can claim cooperative intent, shift group beliefs in a direction that benefits it individually, and then defect without bearing any cost. This is the analogue of the gossip-honesty problem in reputation systems: if communicative acts are not answerable to outcomes, reputational information degrades and cooperation collapses [Számádó et al., 2021]. Term 3 is designed to give communicative acts the weight of interoceptive signals: public statements become answerable to collective outcomes.

Before acting, each member j of group G_t reports a prediction $\hat{g}_t^j \in [0, 1]$ of the group’s cooperation rate. We measure within-group disagreement and consensus prediction error:

$$\bar{g}_t = \frac{1}{|G_t|} \sum_{j \in G_t} \hat{g}_t^j, \quad \bar{c}_t = \frac{1}{|G_t|} \sum_{j \in G_t} \mathbf{1}\{a_t^j = C\}, \quad (6)$$

$$\text{CFE}_t = \frac{1}{|G_t|} \sum_{j \in G_t} D_{\text{KL}}(\text{Bern}(\hat{g}_t^j) \parallel \text{Bern}(\bar{g}_t)) + D_{\text{KL}}(\text{Bern}(\bar{g}_t) \parallel \text{Bern}(\bar{c}_t)), \quad (7)$$

with ε -clipped Bernoulli KL for numerical stability. Every group member receives the same collective penalty:

$$r_{3,t}^k = -\lambda_g \text{CFE}_t. \quad (8)$$

This term is non-decomposable: no agent can reduce CFE_t in isolation. Doing so requires the group to converge on a shared and accurate picture of its own forthcoming behaviour [Tang et al., 2024]. The penalty functions as a form of collective stress sharing [Lyons and Levin, 2024]: when any member’s forecast distorts the group’s self-model, the resulting prediction error is felt by all members equally, just as bioelectric stress in a tissue is distributed across the connected cell network rather than remaining local to the source.

The full reward and training objective.

$$r_t^k = \frac{\pi_t^k + c}{b + c} + \lambda_{\text{tom}} [4q \cos(\varphi_t^k) + \frac{c}{b} \cos(2\varphi_t^k)] - \lambda_g \text{CFE}_t, \quad (9)$$

$$J^k(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=1}^T r_t^k \right]. \quad (10)$$

The three terms operate at increasing scales of self: the payoff term at the individual, the HKB term at the dyad, and the CFE term at the group. All three are active simultaneously from the first update, following GRPO-based post-training practice [DeepSeek-AI, 2025], because in biological populations the pressures that produce cooperation co-evolved rather than activating in sequence. Network reciprocity ($b/c > k$) is activated through the interaction topology rather than as a reward term, and serves as a diagnostic.

4.2 Training

We RL-finetune a pretrained LLM to play iterated Donor’s Games using GRPO (Group Relative Policy Optimisation) [DeepSeek-AI, 2025], which computes advantages group-wise across multiple rollouts of the same prompt and dispenses with an explicit value network. We apply GRPO to LoRA adapters (rank 16, $\alpha = 32$) on a Qwen3-4B base model, with plans to scale to Qwen3.5-27B.

Multi-turn rollout. Each training trajectory is a full iterated game of 3–8 rounds with a single partner whose strategy is held constant for the duration of the game. The model generates a decision each round; the environment computes the partner’s response, the round payoff, and updates the prompt with the running history before the next round. The model must therefore infer its partner’s strategy from observed behaviour across rounds, which is what makes theory-of-mind training possible. Within a game, all three memory stores accumulate: the chat history seen by the model, the per-agent decision histories maintained by the environment, and the HKB coordination window used to compute φ . Across games, everything resets: a new partner, a new scenario, empty histories, and φ reinitialised at $\pi/2$.

217 **Curriculum.** Training proceeds in three stages, each building on the previous stage’s weights:

- 218 1. **Direct and indirect reciprocity** (150 steps). The model sees scenarios with varying re-
219 encounter probability w and reputation visibility q . It learns to cooperate more when w or q
220 is high — basic conditional cooperation.
- 221 2. **Group selection** (150 steps). Multi-agent group scenarios in which individual cooperation
222 affects group survival probability. A group survival bonus at the end of each game propagates
223 credit backward to early cooperation decisions via reward-to-go.
- 224 3. **Abstract naturalistic scenes** (200 steps). The model sees everyday situations and replies
225 with CHOICE: 1 or CHOICE: 2, where the mapping to cooperation or defection is ran-
226 domised each turn. This forces transfer of cooperation instincts from stages 1–2 to naturalis-
227 tic language.

228 The reward for each trajectory is the sum of per-round Donor’s Game payoffs. Full hyperparameters
229 and infrastructure details are given in Appendix ??.

230 4.3 Evaluation

231 We evaluate trained models on the Nowak cooperation benchmark, which sweeps each mechanism’s
232 key parameter and measures how the model’s cooperation rate responds. For each axis, we compute
233 the Spearman rank correlation between the parameter values and mean cooperation rates across
234 multiple seeds. A score of +1 means the model perfectly tracks Nowak’s predicted direction; 0 means
235 it ignores the mechanism entirely. Scores are reported on a four-axis radar chart (direct reciprocity,
236 indirect reciprocity, network reciprocity, group selection). Full benchmark methodology is described
237 in [benchmark reference].

238 **END Warning**

239 5 Preliminary Results

- 240 • **Small models cannot play the game.** Models at 4B parameters cooperate at rates near
241 100% regardless of w , q , or c/b . This unconditional cooperation is not alignment but a
242 failure mode analogous to sycophancy [Sharma et al., 2024]: a model that cannot say no is
243 brittle to exploitation.
- 244 • **Large models can.** Models at 27B+ parameters naturally modulate cooperation in response
245 to Nowak’s mechanisms, without any game-specific finetuning.
- 246 • **Finetuning small models allow them to play the game rationally.** After our training
247 regime, small models acquire conditional play — defecting when cooperation is costly,
248 cooperating when reciprocity or reputation sustain it — and substantially improve their
249 reward.
- 250 • **Transfer to social decision-making.** We evaluate on MACHIAVELLI [Pan et al., 2023], a
251 suite of 134 text-based games tracking ethical violations alongside reward. Models trained
252 with group selection (Term 3) and game payoff (Term 1) reduce ethical violations while
253 maintaining competitive reward. We still must test the impact that the second term has.

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660 **Appendix**

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A Experimental Details

A.1 Models

In our experiments, we RL-finetune a pretrained Qwen3-4B base model with LoRA adapters (rank 16, $\alpha = 32$) using DAPO, a variant of proximal policy optimization that replaces KL regularisation with asymmetric clipping. DAPO uses a GRPO-style advantage estimator with a higher upper clip ratio than lower ($\varepsilon_{\text{high}} = 0.28$ versus $\varepsilon_{\text{low}} = 0.2$), which preserves exploration without the conservatism of a KL penalty. We plan to scale this setup to Qwen3.5-27B in future experiments.

A.2 Environment

We use a Linux-based training environment with CUDA-enabled GPUs for efficient rollout generation and policy optimization. Training is implemented in the `verl` framework with FSDP2 across $8 \times$ A100 40GB GPUs. Rollouts are generated asynchronously via SGLang with tensor parallelism across all 8 GPUs. LoRA adapters are merged into the base weights for rollout inference and unmerged for gradient updates.

All experiments are conducted in an iterated donor’s game environment. In each round, the model interacts with a partner according to the current experimental condition, with the environment controlling whether reputational information is visible, whether partners are re-encountered, whether interactions occur on a structured network, and whether agents are embedded in persistent groups. Opponents are drawn from a fixed strategy pool including always-cooperate, always-defect, tit-for-tat, random, and grudger policies, and each opponent’s strategy remains fixed for the duration of a given trajectory.

A.3 Training Details

Multi-turn rollout. Each training trajectory is a full iterated game of 3–8 rounds. At each round, the model generates an action, the environment computes the partner’s response, calculates the round payoff, and returns the next prompt with updated interaction history. The model must therefore infer its partner’s behavioural pattern from observations accumulated over the trajectory. This multi-turn structure is what makes metacognitive and reciprocity-based training possible.

The multi-turn rollout is implemented as a `verl BaseInteraction` subclass. At each step, the model output is parsed for a `CHOICE` token; the environment then resolves both players’ actions, computes payoffs, and formats the next-round prompt.

Curriculum. Training proceeds in three stages, each initialized from the previous stage’s final weights:

1. **Direct and indirect reciprocity** (150 steps). The model sees scenarios with varying re-encounter probability w and reputation visibility q , and learns basic conditional cooperation.
2. **Group selection** (150 steps). The model is placed in multi-agent group scenarios where individual cooperation affects group survival probability, allowing reward-to-go to propagate collective consequences backward to earlier decisions.
3. **Abstract naturalistic scenes** (200 steps). The model sees everyday scenarios rather than explicit payoff tables or the words “cooperate” and “defect”, and responds with `CHOICE: 1` or `CHOICE: 2`, with the mapping randomised each turn.

Optimization hyperparameters. Table 1 lists the core DAPO configuration. The asymmetric clipping ($\varepsilon_{\text{low}} = 0.2$, $\varepsilon_{\text{high}} = 0.28$) follows the DAPO recommendation: the higher upper clip encourages exploration of cooperative policies that deviate from the reference behaviour, while the lower clip helps prevent instability and policy collapse. We use a learning rate of 10^{-4} after preliminary tuning.

Checkpoint schedule. We use cumulative step counts across the curriculum: stage 1 runs from steps 0–150, stage 2 from 150–300, and stage 3 from 300–500. Each stage resumes from the previous stage’s final checkpoint while resetting the data iterator.

Table 1: DAPO training hyperparameters.

Parameter	Value
Advantage estimator	GRPO
Lower clip ratio (ε_{low})	0.2
Upper clip ratio ($\varepsilon_{\text{high}}$)	0.28
Clip-higher ceiling	10.0
KL penalty	None
Learning rate	1×10^{-4}
Train batch size	16
Generations per prompt	8
Sampling temperature	0.8
Max response length	2048 tokens
LoRA rank	16
LoRA α	32
Target modules	q,k,v,o,gate,up,down_proj

714 A.4 Evaluation Details

715 We evaluate trained models on the Nowak cooperation benchmark, which sweeps the key control
716 parameter for each cooperation mechanism and measures the resulting cooperation rate. For each axis,
717 we compute the Spearman rank correlation between the parameter values and the mean cooperation
718 rate across multiple seeds. A score of +1 indicates perfect alignment with Nowak’s predicted
719 direction, whereas a score of 0 indicates no systematic response.

720 The benchmark reports four axes: direct reciprocity, indirect reciprocity, network reciprocity, and
721 group selection. Direct reciprocity is evaluated by sweeping the re-encounter probability w ; indirect
722 reciprocity by sweeping reputation visibility q ; network reciprocity by varying local graph degree k
723 relative to the benefit-cost ratio b/c ; and group selection by varying the balance between within-group
724 and between-group competition. Scores are summarized in a four-axis radar plot in the main paper.

725 B Limitations

726 Our current experiments are limited in several ways. First, we instantiate kin selection only implicitly
727 through shared structure and do not yet implement it as a dedicated training pressure, even though it
728 is one of Nowak’s original five rules. Second, our theory-of-mind signal is operationalized through
729 coordination dynamics rather than explicit belief-state modeling, which captures one important
730 aspect of social cognition but not the full space of metacognitive representations. Third, the current
731 experiments are conducted on a relatively small base model and within stylized donor’s game
732 environments; it remains an open question how well the same training pressures transfer to larger
733 models and more naturalistic multi-agent domains. Finally, several benchmark references and
734 empirical result summaries remain placeholders in the current draft and must be completed before
735 submission.